Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	LTVIP2026TMIDS80982
Project Name	Prosperity Prognosticator: Machine Learning for Startup Success Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a developer, I can collect historical startup dataset for success prediction.	2	High	All
Sprint-1	Data Collection	USN-2	As a developer, I can load dataset into the system for processing.	1	High	Member-1
Sprint-1	Data Preparation	USN-3	As a developer, I can handle missing values in the dataset.	3	High	Member-2
Sprint-1	Data Preparation	USN-4	As a developer, I can clean and preprocess startup data.	3	High	Member-3
Sprint-1	Data Preparation	USN-5	As a developer, I can handle imbalanced startup outcome data.	5	High	Member-1
Sprint-2	Model Development	USN-6	As a developer, I can split dataset into training and testing sets.	2	High	Member-2

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Model Development	USN-7	As a developer, I can train ML model for startup success prediction.	5	High	Member-1
Sprint-2	Model Evaluation	USN-8	As a developer, I can evaluate model using accuracy, precision, recall and F1-score.	3	Medium	Member-3
Sprint-2	Backend Development	USN-9	As a developer, I can integrate trained model with Flask backend.	5	High	Member-2
Sprint-2	Frontend Development	USN-10	As a user, I can enter startup details and view prediction result (Acquired / Closed).	3	High	Member-3

Project Tracker, Velocity & Burndown Chart: (4 Marks) :

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	14	6 Days	01 Feb 2025	06 Feb 2025	14	06 Feb 2025
Sprint-2	18	6 Days	08 Feb 2025	13 Feb 2025	18	13 Feb 2025

Velocity Calculation

Velocity Formula:

$$Velocity = TotalStoryPoints/NumberofSprints$$

Velocity = 32 / 2 = 16

Team Velocity = **16 Story Points per Sprint**

Average Velocity per Day

If Sprint Duration = 6 Days
Velocity per Sprint = 16 Points

$$AverageVelocityperDay = Velocity/SprintDuration$$

Average Velocity per Day = 16 / 6 ≈ 2.67

👉 Team completes approximately **2.67 Story Points per Day**

Burndown Chart

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

For the **Startup Success Prediction using Machine Learning** project, the burndown chart shows:

• X-axis → Sprint Days (1 to 6)
• Y-axis → Remaining Story Points
• Ideal Line → Planned reduction of story points
• Actual Line → Real progress made by the team